

# Simplicial (Complex) Convolutional NNs

[Yang et al. 2025 TMLR]

# Simplicial CNNs

Edge-only interactions, Chebnet for edges

Linear

$$\mathbf{H} := \mathbf{H}(\mathbf{L}_d, \mathbf{L}_u; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \sum_{k=0}^{K_d} \alpha_k \mathbf{L}_d^k + \sum_{k=0}^{K_u} \beta_k \mathbf{L}_u^k$$

Non-Linear

